

CG3207 – COMPUTER ARCHITECTURE : QUIZ – I

Duration : 15 minutes

Total marks : 5

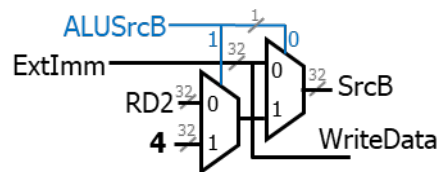
Instructions

Where relevant, assume RV32I ISA and the RISC-V microarchitecture covered in CG3207 Chapter 3B. Each question carries 1 mark (no partial marks). Write your answers only on this side of the paper.

1. The number of flip-flops inferred by the following VHDL / Verilog code, assuming all the identifiers are 1-bit is _____.

VHDL	Verilog
<pre> Z <= X and Y; process (P, Q, Z) variable A,C,E : std_logic := '0'; begin A := P xor Q; C := P xor Z; D <= C; E := C and A; B <= E; end process; </pre>	<pre> reg A,C,E; assign Z = X & Y; always @ (P or Q or Z) begin A = P ^ Q; C = P ^ Z; D <= C; E = C & A; B <= E; end </pre>

2. The connections to the single-cycle processor are modified to be as follows. The ALUSrcB[1:0] corresponding to addi instruction is _____.



3. Following the modification mentioned in Q.2 above, the instruction that will be executed incorrectly even if ALUSrcB[1:0] is asserted correctly is
- lw
 - sw
 - beq
 - auipc
 - add
 - jal (with support for link)
4. Which of the following instructions is/are **legal** in RV32I? (multiple response – choose all applicable options).
- add t1, t2, 2
 - sltiu t1, t2, 0x0000FFF
 - mulh t1, t2, t3, t4
 - slti t1, t2, 0xFFFFFFFF
 - subi t1, t2, 0xFF
5. The value in x7 after executing the code below is _____ (you can provide your answer in any radix, with an appropriate C or Verilog style prefix if not decimal).

```

addi x5, x0, 1
addi x6, x0, -1
bgeu x5, x6, skip
addi x7, x0, 10
skip: addi x7, x7, 20

```

A								
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CG3207 – COMPUTER ARCHITECTURE : QUIZ – II**Duration : 15 minutes****Total marks : 5****Instructions**

Each question carries 1 mark (no partial marks). Write your answers only on this side of the paper.

1. A particular benchmarking scheme is similar to SPECINT2006, except that it uses 3 programs to measure performance. The execution times of these programs on the machine under test are 10 ns, 12 ns, and 20 ns, respectively. The corresponding runtimes on the reference machine are 40 ns, 60 ns, and 100 ns, respectively. Assuming that the I/O and other overheads are negligible; the benchmark score for the machine under test is _____ (no need to simplify the result).
2. In an *optimised* design, the number of 1-bit 2 to 1 multiplexers required to convert a circuit capable of doing logical right shifts (on 32-bit inputs) to be able to handle both arithmetic and logical right shifts is _____.
3. A double precision floating point number (1 sign bit, 11 exponent bits, 52 fraction bits) can represent numbers in the range of (approximately)
 - a) 2^{-2048} to 2^{2048}
 - b) 2^{-1024} to 2^{1024}
 - c) -2^{2048} to 2^{2048}
 - d) -2^{1024} to 2^{1024}
4. The cost of a sequential multiplier _____ (assume that the adder used is ripple carry) with the number bits.
 - a) increases linearly
 - b) increases logarithmically
 - c) increases exponentially
 - d) remains unchanged
5. The result of multiplication of two 2^n -bit numbers may require up to _____ bits.
 - a) 2^{2n}
 - b) $n+1$
 - c) $2n+1$
 - d) 2^{n+1}

CG3207 – COMPUTER ARCHITECTURE : QUIZ – 3**Duration : 15 minutes****Total marks : 10****Instructions**

All questions carry 1 mark each. Partial marks will not be awarded. The other side of this sheet can be used for rough work, and will not be graded.

1. Executing the following instructions

```
lw s4, 8(s2)
```

```
sw s2, 0(s4)
```

will cause _____ (enter A/B/C/D/E), assuming that the processor designer has taken into account the presence of a forwarding circuitry to deal with the memory-to-memory copy scenario while designing the 'load and use' stall circuitry.

- A) No stall, and forwarding from M to E
- B) No stall, and forwarding from W to M
- C) Stall, and forwarding from M to E
- D) Stall, and forwarding from W to E
- E) Stall, and forwarding from W to M

2. The expression for lwStall in the case above should be [$\sim$ denotes bitwise not]

lwStall = ((rs1D == rdE) | ((rs2D == rdE) & ($\sim$ _____))) & MemtoRegE & ~~MemtoWriteE~~

3. Assuming that the branch in a 5-stage pipelined processor* is committed in the **W** stage, and that the ISA specifies the branch delay slot to have 2 instructions, the number of instructions that will need to be flushed when the branch is taken is _____.

*Assume it is similar to the RISC-V pipelined processor we saw in lectures with the differences as mentioned above. The official RISC-V ISA does not include/specify any branch delay slots.

4. Perform register renaming to eliminate storage conflicts for the instructions below. Use a, b, c, .. to fill in the blanks below, using higher alphabets / names only when necessary. For example, x1b should be used only if x1a is not sufficient to eliminate the storage conflict.

```
lw x4a, 8(x3a)
sw x3__, (x4__)
add x3__, x3__, x4__
add x4__, x4__, x3__
```

5. Statement X: The `volatile` qualifier is necessary *and* sufficient to ensure that there are no issues related to the ordering of memory operations when performing memory-mapped input/output (MMIO).

Statement Y: In Programmed I/O (PIO), the processor does not access the memory directly, whereas in Direct Memory Access (DMA), the processor accesses the memory directly.

- A) Statement X is true, statement Y is false.
- B) Statement Y is true, statement X is false.
- C) Both statements are true.
- D) Both statements are false.